

NexRay AI

Medical Assistant Platform • Clinical Decision Support Report

19 July 2026 | 10:32 PM

Session ID: 8	Modules Used: Symptom Checker	Date: 19 July 2026 10:32 PM
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Symptom Analysis

Primary Condition	Tuberculosis (TB)
Likelihood	High
Assessment	The combination of persistent cough for 2 weeks, night sweats, weight loss, and fatigue is classic for pulmonary tuberculosis. This presentation is common in West Africa where TB prevalence remains significant. The chronic nature of symptoms and systemic manifestations strongly suggest TB over acute respiratory infections.

■ Urgency Level: Urgent — Same day attention required

Recommended Tests

- Sputum smear microscopy (Ziehl-Neelsen staining) - at least 2 samples
- Chest X-ray to assess for typical TB infiltrates
- GeneXpert MTB/RIF test for rapid TB and rifampicin resistance detection
- HIV testing (important co-infection consideration in West Africa)
- Full blood count and liver function tests

Suggested Treatment

- Initiate anti-TB therapy after confirmatory testing using first-line regimen (HRZE for 2 months, followed by HR for 4 months) following national TB guidelines
- Nutritional support and vitamin B6 supplementation during TB treatment
- Contact tracing for household and close contacts
- Monitor for treatment adherence and side effects regularly

Next Steps

- Isolate patient pending confirmation to prevent transmission
- Refer to TB specialist or TB unit for case management if available
- Register case with national TB programme
- Arrange baseline assessment including HIV status before starting treatment

Summary: High suspicion for pulmonary tuberculosis based on classic chronic presentation with systemic symptoms. Urgent confirmatory testing required, particularly GeneXpert MTB/RIF and sputum microscopy. Patient should be isolated pending results and managed according to national TB treatment guidelines. HIV co-infection screening is essential.

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